The Robert M. Buchan Department of Mining

Queen's University

MNTC 313 – Assignment 3

A common situation dealt with in engineering and physics is that of projectile motion. The equation for the height of a projectile, y [m], is:

$$y = -\left(\frac{1}{2}\right) g t^2 \; + \; V_0 \sin(\Theta_0)\, t + y_0,$$

where

g = Acceleration due to gravity: 9.81 m/s^2

t = Time (seconds)

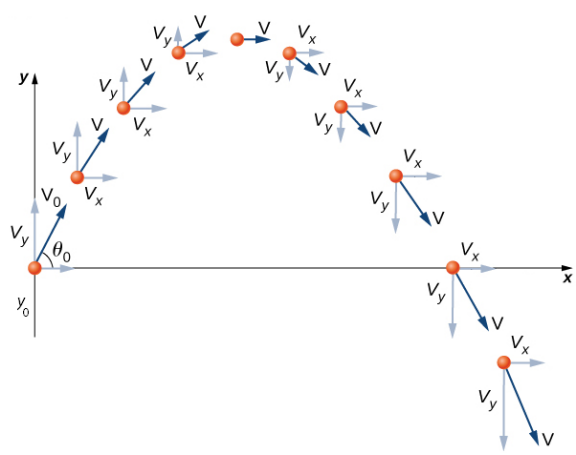
V_0 = Initial Velocity of Projectile (m/s)

Θ_0 = Launch Angle (degrees)

y_0= Initial height of the projectile (m)

The equation for the vertical velocity of the projectile, x [m], is:

$$V_y \; = \; -gt \; + \; V_0 \sin(\Theta_0).$$



Write a program using C/C++ to determine the projectile motion (height, vertical velocity) at a certain time.

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Your code should then employ the following logical conditions:

- If time is greater or equal than zero, first print "Projectile released"
  - Print the position and velocity value with the following format:
    "Position: xx.xx meters"
    "Velocity value: xx.xx meters/second"
  - If the projectile has a vertical velocity greater or equal than zero, print "Projectile ascending"
    - If the vertical position of the projectile is greater or equal than the initial vertical position, print "Projectile above initial position"
  - Or, if the projectile has a negative vertical velocity, print "Projectile descending"
    - If the vertical position of the projectile is greater or equal than the initial vertical position, print "Projectile above initial position"
    - Otherwise, print "Projectile below initial position"
- Or, if time is lower than zero, print "Projectile not released"

Your output must look like:

```
10,10,45,10
Projectile released
Position: -409.82 meters
Velocity value: -91.03 meters/second
Projectile descending
Projectile below initial position
```

And for negative time must look like:

```
-10,10,45,10
Projectile not released
```

Note: **The first line in each example above contains sample inputs passed via the scanf() statement. Your program should function properly for any inputs.** A starter code file has been provided on OnQ. Download the file, and copy and paste its contents into a new .c/.cpp file in your IDE. Use pi = 3.14.

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